Ketones - Blerd

AUTHOR
BDH

Data Set Up

```r
# 1.
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   4.0.1     ✔ tibble    3.2.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.0.4
── Conflicts ──────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to b
```

```r
library(car)
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

```r
library(gt)
library(modelsummary)
library(emmeans)
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```r
library(modelbased)
```

```
Warning: package 'modelbased' was built under R version 4.4.3
```

```r
library(multcomp)
```

```
Loading required package: mvtnorm

Attaching package: 'mvtnorm'

The following object is masked from 'package:modelbased':

    standardize

Loading required package: survival
Loading required package: TH.data
Loading required package: MASS

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select


Attaching package: 'TH.data'

The following object is masked from 'package:MASS':

    geyser
```

```r
library(ggpubr)
library(cowplot)
```

```

Attaching package: 'cowplot'

The following object is masked from 'package:ggpubr':

    get_legend

The following object is masked from 'package:gt':

    as_gtable

The following object is masked from 'package:lubridate':

    stamp
```

```r
library(lme4)
```

```
Warning: package 'lme4' was built under R version 4.4.3

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```r
library(lmerTest)
```

```

Attaching package: 'lmerTest'

The following object is masked from 'package:lme4':

    lmer

The following object is masked from 'package:stats':

    step
```

```r
library(outliers)
library(DHARMa)
```

```
This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package
```

```r
library(statmod)
library(readr)
library(dplyr)
library(EnvStats)
```

```

Attaching package: 'EnvStats'

The following object is masked from 'package:Matrix':

    print

The following object is masked from 'package:MASS':

    boxcox

The following object is masked from 'package:car':

    qqPlot

The following objects are masked from 'package:stats':

    predict, predict.lm

The following object is masked from 'package:base':

    print.default
```

```r
library(performance)
```

```
Warning: package 'performance' was built under R version 4.4.3
```

```r
library(see)
library(patchwork)
```

```

Attaching package: 'patchwork'

The following object is masked from 'package:cowplot':

    align_plots

The following object is masked from 'package:MASS':

    area
```

```r
# 2.
setwd("/Users/ben/Desktop/Blerd/Data Analysis")

# 3. Load your data
datum = read.csv("Blerd.csv")
datumlong = read.csv("BlerdData_long.csv")

# 4. Ensure categorical variables are properly set
datum$SexF = as.factor (datum$Sex)
datum$TreatmentF = as.factor(datum$Treatment)
datum$TimeF = as.factor(datum$Time)
datum$IncubatorF = as.factor (datum$Incubator)
datum$SideF = as.factor (datum$Side)
datum$PostKetones = as.numeric (datum$PostKetones)
```

```
Warning: NAs introduced by coercion
```

```r
datum$PreKetones = as.numeric (datum$PreKetones)

datumlong$SexF = as.factor (datumlong$Sex)
datumlong$TreatmentF = as.factor(datumlong$Treatment)
datumlong$TimepointF = as.factor(datumlong$Timepoint)
datumlong$Ketones = as.numeric (datumlong$Ketones)
```

```
Warning: NAs introduced by coercion
```

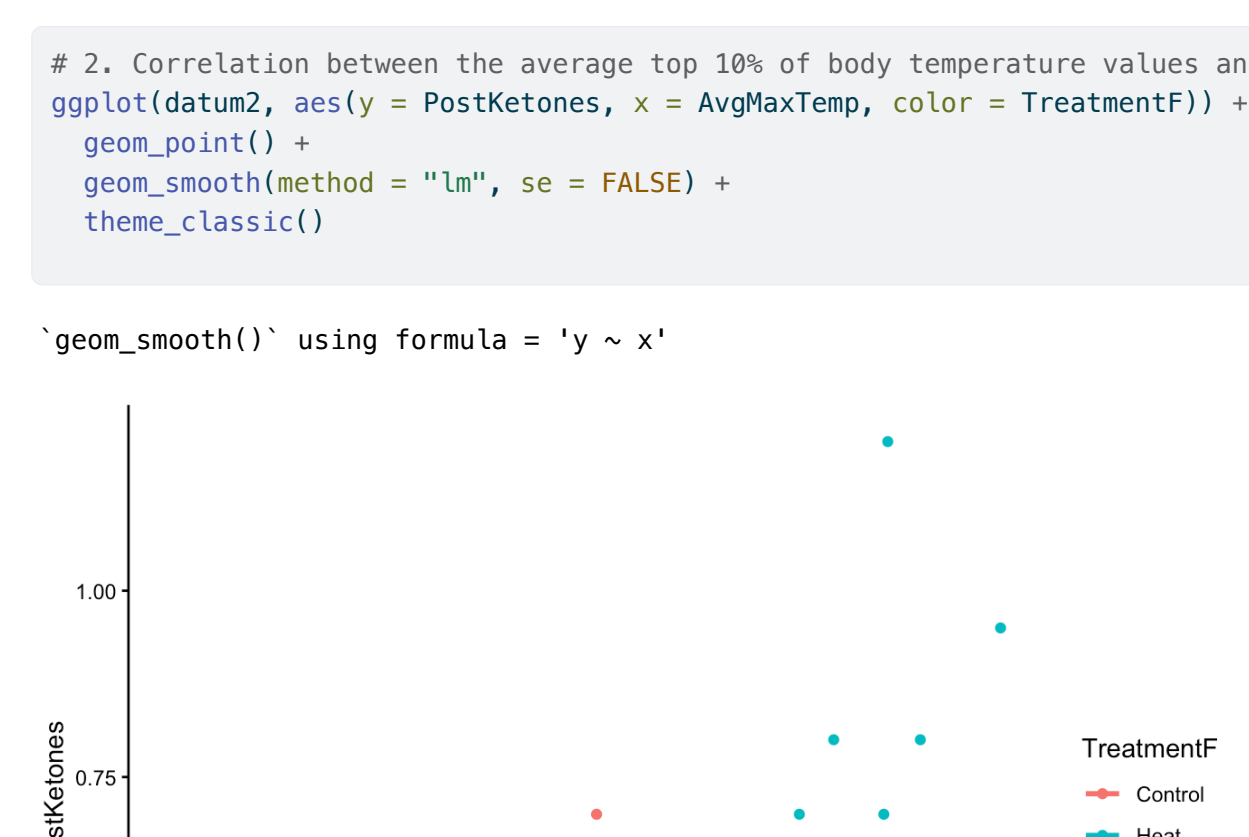
```r
datumlong$Glucose = as.numeric (datumlong$Glucose)
```

```
Warning: NAs introduced by coercion
```

```r
# 5. Subset data so there are no dead individuals included
datum1 = datum %>%
  filter(!Died %in% c("Died", "PIT.Death"))

datumlong1 = datumlong %>%
  filter(!Died %in% c("Died", "PIT.Death"))
datumlong1 = datumlong1[-47, ]

# 6. Take out any of the blank cells that cause problems when graphing
datum2 = datum1 %>%
  filter(PostKetones != "", PostKetones != "Dead", AvgMaxTemp != "", AvgMaxTemp != "Dead
datum2$TreatmentF = as.factor(datum2$Treatment)

datumlong2 = datumlong1 %>%
  filter(Ketones != "", Ketones != "Dead")
```

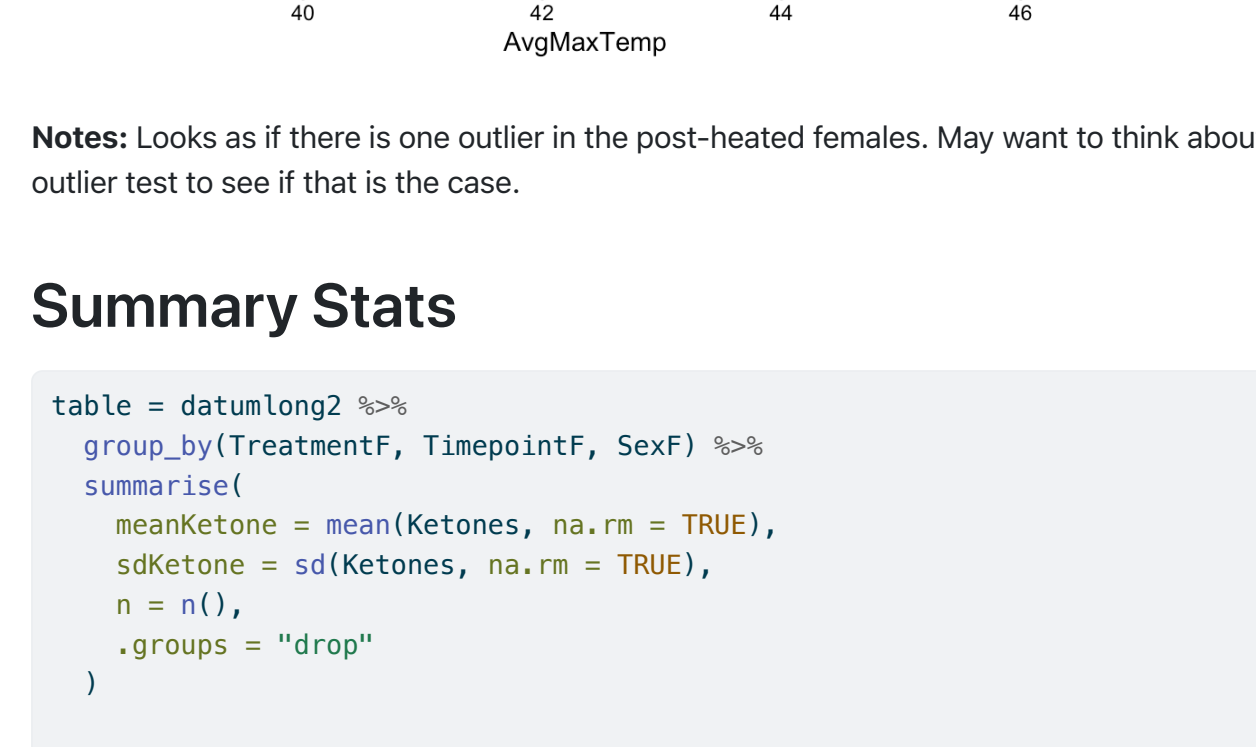
Initial Data Viz

```r
# 1. Ketones across timepoints and relationship to treatment and sex
ggplot(datumlong2, aes(x = Timepoint, y = Ketones, color = SexF, shape = TreatmentF)) +
  geom_point(
    position = position_jitterdodge(
      jitter.width = 0.2,
      dodge.width = 0.5
    )
  ) +
  theme_classic()
```



```r
# 2. Correlation between the average top 10% of body temperature values and post ketones
ggplot(datum2, aes(y = PostKetones, x = AvgMaxTemp, color = TreatmentF)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_classic()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Notes: Looks as if there is one outlier in the post-heated females. May want to think about running an outlier test to see if that is the case.

Summary Stats

```r
table = datumlong2 %>%
  group_by(TreatmentF, TimepointF, SexF) %>%
  summarise(
    meanKetone = mean(Ketones, na.rm = TRUE),
    sdKetone = sd(Ketones, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

print(table)
```

```
  TreatmentF TimepointF SexF meanKetone   sdKetone  n
1    Control       Post    F  0.3694444 0.08598716 18
2    Control       Post    M  0.5361111 0.16873714 18
3    Control        Pre    F  0.4525000 0.10818478 20
4    Control        Pre    M  0.4500000 0.11592023 17
5       Heat       Post    F  0.5472222 0.21792620 18
6       Heat       Post    M  0.5411765 0.18644310 17
7       Heat        Pre    F  0.4522727 0.11900107 22
8       Heat        Pre    M  0.4214286 0.12606688 21
```

Analysis - Ketones across time points

We started under a linear mixed effects model framework to test for the effects of treatment (control or heated), sex (male or female), and time point (pre or post) on ketone levels. Because we have two samples for each individual, we included that as a random effect (1|BirdID).

```r
#Running the models

#  1. Start with global linear mixed effects model to assess ketone levels
results = lmer(Ketones ~ TreatmentF * SexF * TimepointF + (1|ID),
               data = datumlong2, na.action = na.omit)
summary(results)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ TreatmentF * SexF * TimepointF + (1 | ID)
   Data: datumlong2

REML criterion at convergence: -124.3

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.0195 -0.5865 -0.0332  0.4607  4.2171

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.004107 0.06409
 Residual             0.017093 0.13074
Number of obs: 151, groups:  ID, 82

Fixed effects:
                                     Estimate Std. Error        df t value
(Intercept)                           0.36885    0.03425 139.70685  10.768
TreatmentFHeat                        0.18312    0.04841 140.38467   3.783
SexFM                                 0.16777    0.04846 139.27952   3.462
TimepointFPre                         0.08365    0.04269  66.61056   1.959
TreatmentFHeat:SexFM                 -0.17612    0.06897 140.24331  -2.554
TreatmentFHeat:TimepointFPre         -0.18334    0.05986  68.23094  -3.063
SexFM:TimepointFPre                  -0.16768    0.06173  67.90715  -2.716
TreatmentFHeat:SexFM:TimepointFPre    0.14519    0.08618  68.98804   1.685
                                    Pr(>|t|)
(Intercept)                          < 2e-16 ***
TreatmentFHeat                      0.000229 ***
SexFM                               0.000713 ***
TimepointFPre                       0.054264 .
TreatmentFHeat:SexFM                0.011730 *
TreatmentFHeat:TimepointFPre        0.003134 **
SexFM:TimepointFPre                 0.008364 **
TreatmentFHeat:SexFM:TimepointFPre 0.096564 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TrFH:SFM TFH:TF SFM:TF
TreatmntFHt -0.708
SexFM       -0.707  0.500
TimepontFPr -0.662  0.468  0.468
TrtmnFH:SFM  0.497 -0.702 -0.703 -0.329
TrtmnFH:TFP  0.472 -0.673 -0.334 -0.713  0.473
SxFM:TmpnFP  0.458 -0.324 -0.645 -0.692  0.453    0.493
TFH:SFM:TFP -0.328  0.468  0.462  0.495 -0.665   -0.695 -0.716
```

```r
Anova(results, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
                              Chisq Df Pr(>Chisq)
(Intercept)                115.9523  1  < 2.2e-16 ***
TreatmentF                  14.3107  1  0.0001550 ***
SexF                        11.9833  1  0.0005368 ***
TimepointF                   3.8389  1  0.0500766 .
TreatmentF:SexF              6.5209  1  0.0106616 *
TreatmentF:TimepointF        9.3826  1  0.0021905 **
SexF:TimepointF              7.3789  1  0.0065995 **
TreatmentF:SexF:TimepointF   2.8382  1  0.0920444 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 2. The three way interaction was not significant, so it was removed and all two way in
results2 = lmer(Ketones ~ TreatmentF * SexF + TreatmentF * TimepointF +
                SexF * TimepointF + (1|ID), data = datumlong2, na.action = na.omit)
summary(results2)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ TreatmentF * SexF + TreatmentF * TimepointF + SexF *
    TimepointF + (1 | ID)
   Data: datumlong2

REML criterion at convergence: -124.5

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.8592 -0.6084 -0.1048  0.5028  4.3484

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.003825 0.06185
 Residual             0.017584 0.13261
Number of obs: 151, groups:  ID, 82

Fixed effects:
                              Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                    0.38781    0.03250 132.90207  11.934  < 2e-16
TreatmentFHeat                 0.14461    0.04292 121.42986   3.369  0.00101
SexFM                          0.13000    0.04311 116.25455   3.016  0.00315
TimepointFPre                  0.04797    0.03760  68.54610   1.276  0.20631
TreatmentFHeat:SexFM          -0.09864    0.05145  71.74737  -1.917  0.05919
TreatmentFHeat:TimepointFPre  -0.11291    0.04365  70.14413  -2.587  0.01177
SexFM:TimepointFPre           -0.09332    0.04366  70.25753  -2.137  0.03605

(Intercept)                  ***
TreatmentFHeat               **
SexFM                        **
TimepointFPre
TreatmentFHeat:SexFM         .
TreatmentFHeat:TimepointFPre *
SexFM:TimepointFPre          *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TFH:SF TFH:TF
TreatmntFHt -0.663
SexFM       -0.662  0.360
TimepontFPr -0.614  0.311  0.313
TrtmnFH:SFM  0.393 -0.590 -0.595  0.001
TrtmnFH:TFP  0.363 -0.553 -0.020 -0.591  0.020
SxFM:TmpnFP  0.341  0.018 -0.513 -0.556 -0.045 -0.008
```

```r
Anova(results2, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
                         Chisq Df Pr(>Chisq)
(Intercept)           142.4147  1  < 2.2e-16 ***
TreatmentF             11.3501  1  0.0007545 ***
SexF                    9.0933  1  0.0025654 **
TimepointF              1.6278  1  0.2020056
TreatmentF:SexF         3.6757  1  0.0552126 .
TreatmentF:TimepointF   6.6910  1  0.0096903 **
SexF:TimepointF         4.5685  1  0.0325660 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# 3. The Treatment*Sex interaction was not significant, so it was removed from the model
results3 = lmer(Ketones ~ SexF * TimepointF + TreatmentF * TimepointF +
                (1|ID), data = datumlong2, na.action = na.omit)
summary(results3)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ SexF * TimepointF + TreatmentF * TimepointF + (1 |      ID)
   Data: datumlong2

REML criterion at convergence: -125

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.7926 -0.5757 -0.1255  0.5126  4.4203

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.004181 0.06466
 Residual             0.017670 0.13293
Number of obs: 151, groups:  ID, 82

Fixed effects:
                              Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                    0.41229    0.03019 141.88843  13.659  < 2e-16
SexFM                          0.08081    0.03501 142.26411   2.308  0.02243
TimepointFPre                  0.04810    0.03770  68.02433   1.276  0.20640
TreatmentFHeat                 0.09632    0.03501 142.24325   2.751  0.00671
SexFM:TimepointFPre           -0.09696    0.04375  69.78822  -2.216  0.02994
TimepointFPre:TreatmentFHeat  -0.11161    0.04377  69.61965  -2.550  0.01298

(Intercept)                  ***
SexFM                        *
TimepointFPre
TreatmentFHeat               **
SexFM:TimepointFPre          *
TimepointFPre:TreatmentFHeat *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) SexFM  TmpnFP TrtmFH SFM:TF
SexFM       -0.579
TimepontFPr -0.664  0.388
TreatmntFHt -0.581  0.014  0.384
SxFM:TmpnFP  0.388 -0.667 -0.556 -0.011
TmpntFP:TFH  0.383 -0.011 -0.591 -0.667 -0.007
```

```r
Anova(results3, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
                         Chisq Df Pr(>Chisq)
(Intercept)           186.5620  1  < 2.2e-16 ***
SexF                    5.3280  1   0.020985 *
TimepointF              1.6274  1   0.202058
TreatmentF              7.5694  1   0.005937 **
SexF:TimepointF         4.9117  1   0.026675 *
TimepointF:TreatmentF   6.5021  1   0.010775 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Use AIC to test which one of these models has the best fit
AIC(results, results2, results3)
```

```
         df       AIC
results  10 -104.2897
results2  9 -106.5421
results3  8 -108.9974
```

```r
#This is the global model, use 'results3' for testing the assumptions

results_final = lmer(Ketones ~ SexF * TimepointF + TreatmentF * TimepointF +
                (1|ID), data = datumlong2, na.action = na.omit)
summary(results_final)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ SexF * TimepointF + TreatmentF * TimepointF + (1 |      ID)
   Data: datumlong2

REML criterion at convergence: -125

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.7926 -0.5757 -0.1255  0.5126  4.4203

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.004181 0.06466
 Residual             0.017670 0.13293
Number of obs: 151, groups:  ID, 82

Fixed effects:
                              Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                    0.41229    0.03019 141.88843  13.659  < 2e-16
SexFM                          0.08081    0.03501 142.26411   2.308  0.02243
TimepointFPre                  0.04810    0.03770  68.02433   1.276  0.20640
TreatmentFHeat                 0.09632    0.03501 142.24325   2.751  0.00671
SexFM:TimepointFPre           -0.09696    0.04375  69.78822  -2.216  0.02994
TimepointFPre:TreatmentFHeat  -0.11161    0.04377  69.61965  -2.550  0.01298

(Intercept)                  ***
SexFM                        *
TimepointFPre
TreatmentFHeat               **
SexFM:TimepointFPre          *
TimepointFPre:TreatmentFHeat *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) SexFM  TmpnFP TrtmFH SFM:TF
SexFM       -0.579
TimepontFPr -0.664  0.388
TreatmntFHt -0.581  0.014  0.384
SxFM:TmpnFP  0.388 -0.667 -0.556 -0.011
TmpntFP:TFH  0.383 -0.011 -0.591 -0.667 -0.007
```

```r
Anova(results_final, type = "III")
```

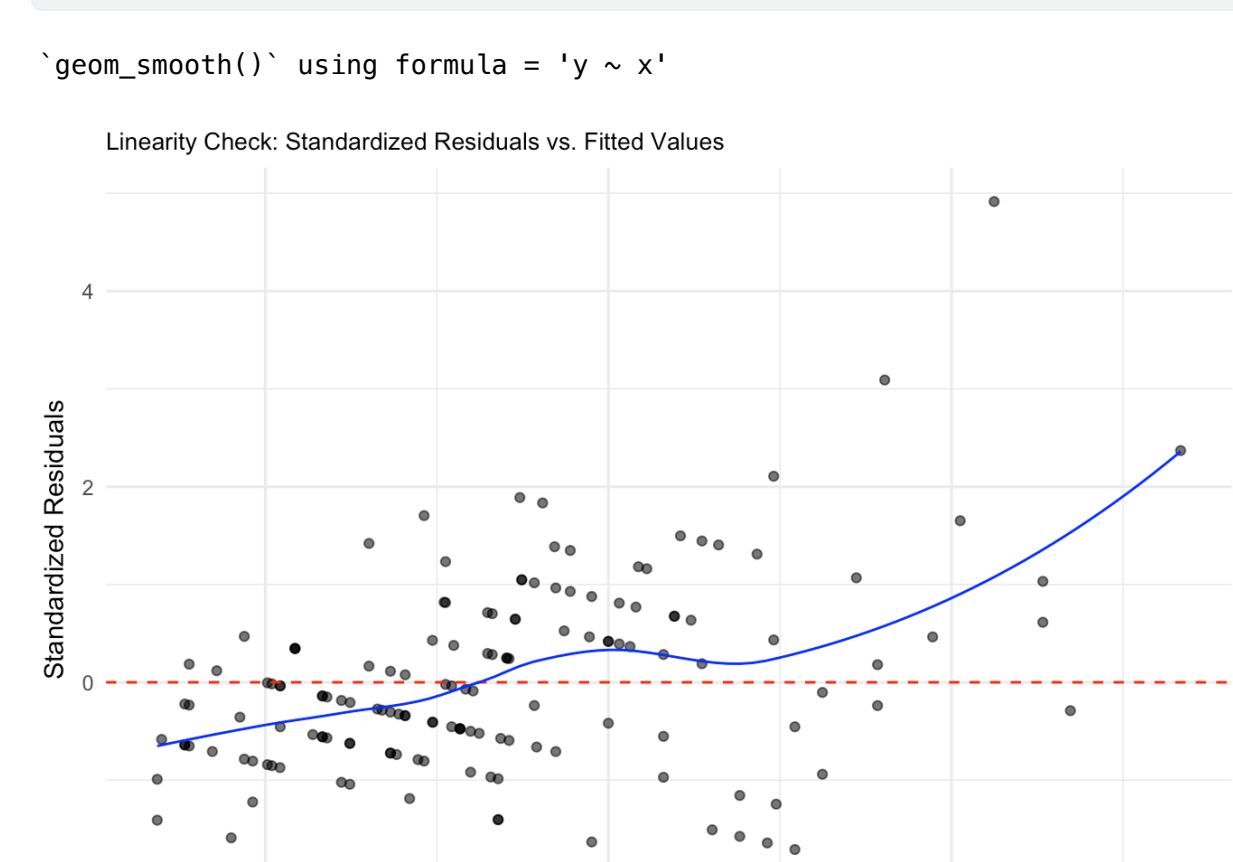
```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
                         Chisq Df Pr(>Chisq)
(Intercept)           186.5620  1  < 2.2e-16 ***
SexF                    5.3280  1   0.020985 *
TimepointF              1.6274  1   0.202058
TreatmentF              7.5694  1   0.005937 **
SexF:TimepointF         4.9117  1   0.026675 *
TimepointF:TreatmentF   6.5021  1   0.010775 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Assumptions of the model


# 1. Check the VIF of this model
vif(results_final)
```

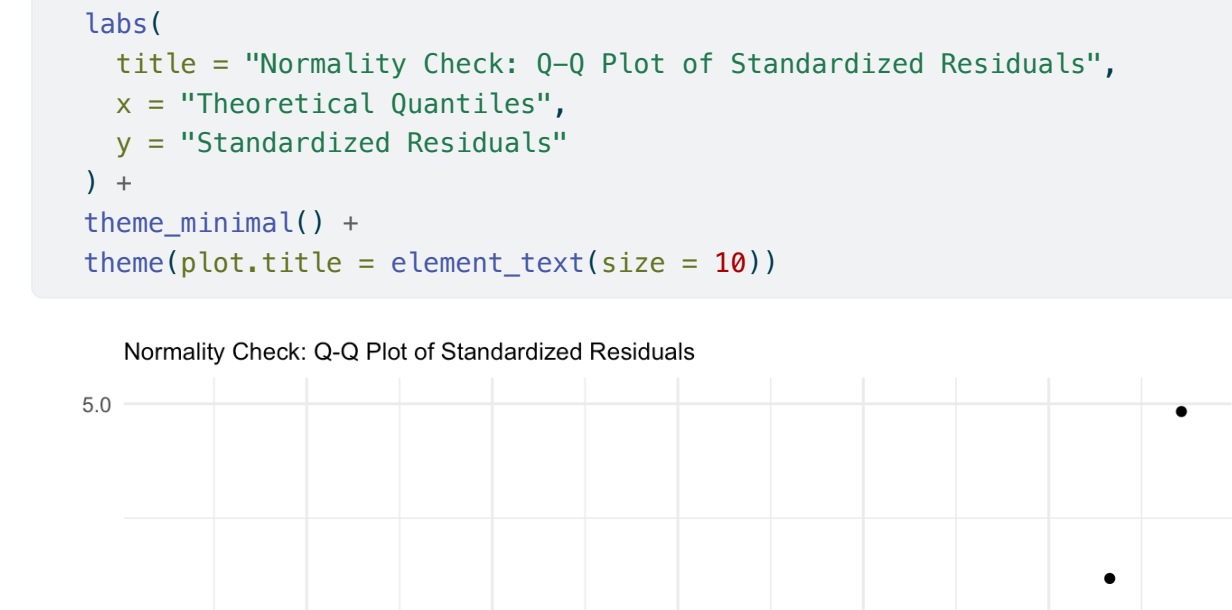
```
                 SexF            TimepointF            TreatmentF
             1.804651              2.976805              1.804127
      SexF:TimepointF TimepointF:TreatmentF
             2.711204              2.936115
```

```r
  #VIF = 1: No multicollinearity
  #VIF between 1 and 5: Generally considered acceptable or moderate multicollinearity
  #VIF > 5 (or > 10): Indicates potentially high multicollinearity

# 2. Check other assumptions of a linear model

# --- Extract Residuals and Fitted Values ---
#1. Extract the data frame used by the model
datum_clean <- model.frame(results_final)

# 2. Calculate and assign the fitted and residual values to this clean data frame
datum_clean$fitted <- fitted(results_final)
datum_clean$raw_resid <- residuals(results_final)
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)

# 3. Use the studentized residuals method from the 'lme4' utility functions
# Although 'rstandard' doesn't work, 'residuals' is robust.
# If you really need a standardized residual, use this function:
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)


# --- 2. Create the Plots ---

# A. Linearity and Homoscedasticity Plot (Residuals vs. Fitted)
ggplot(datum_clean, aes(x = fitted, y = std_resid)) +
  geom_point(alpha = 0.6) +
  # Add the ideal reference line at y=0
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  # Add a LOESS smoothing line (red line in performance)
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Linearity Check: Standardized Residuals vs. Fitted Values",
    x = "Fitted Values",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

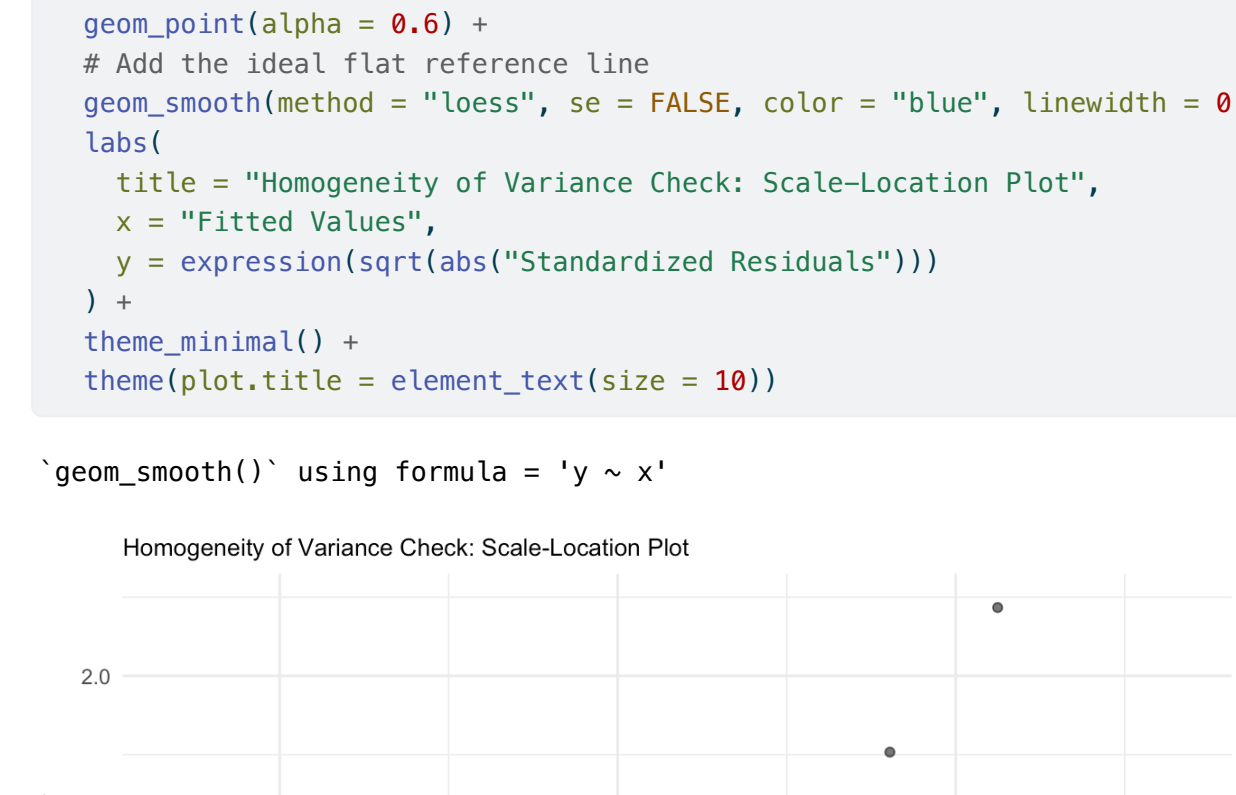


```
# B. Normality Check (Quantile-Quantile Plot)
ggplot(datum_clean, aes(sample = std_resid)) +
  # Create the Q-Q plot
  stat_qq() +
  # Add the reference line (the ideal normal distribution line)
  stat_qq_line(color = "red") +
  labs(
    title = "Normality Check: Q-Q Plot of Standardized Residuals",
    x = "Theoretical Quantiles",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

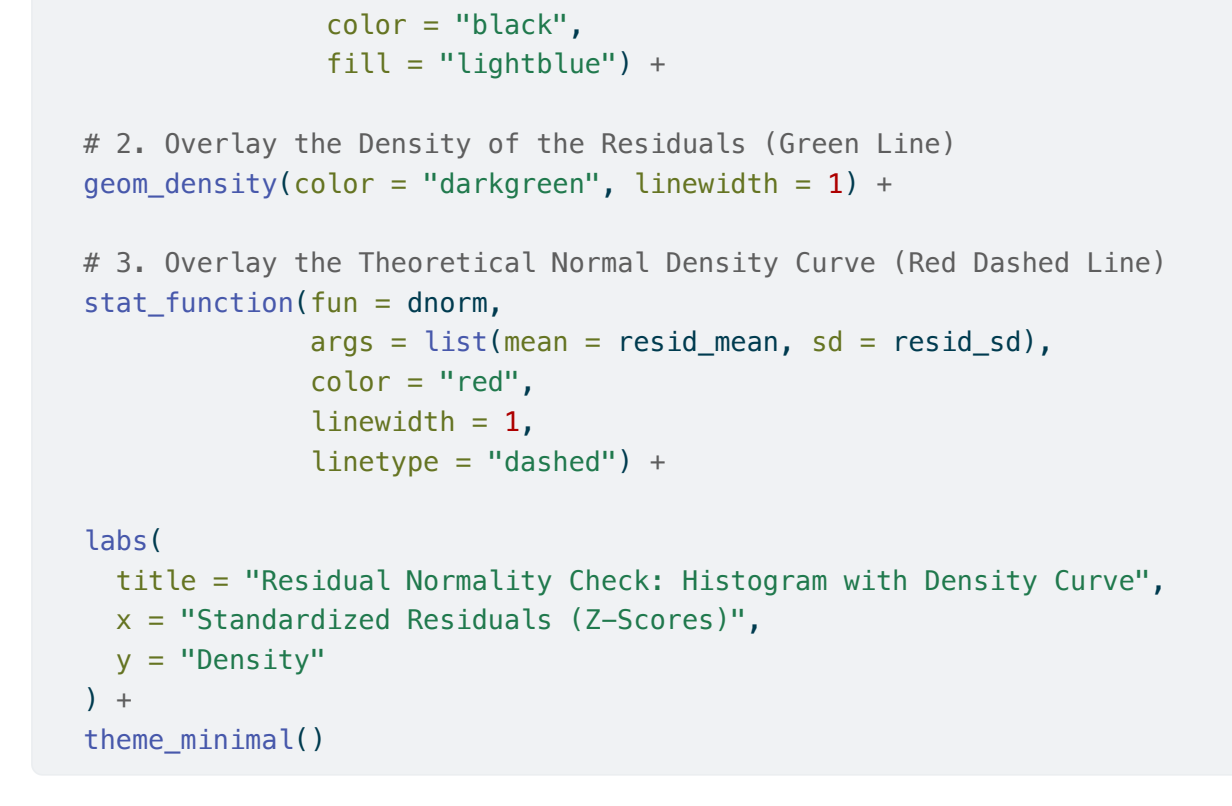


```
# C. Scale-Location Plot (Homogeneity of Variance Check)
# This uses the square root of the absolute standardized residuals
ggplot(datum_clean, aes(x = fitted, y = sqrt(abs(std_resid)))) +
  geom_point(alpha = 0.6) +
  # Add the ideal flat reference line
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Homogeneity of Variance Check: Scale-Location Plot",
    x = "Fitted Values",
    y = expression(sqrt(abs("Standardized Residuals")))
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
#D. Histogram of residuals
# Calculate mean and standard deviation of residuals for the density curve
resid_mean <- mean(datum_clean$std_resid, na.rm = TRUE)
resid_sd <- sd(datum_clean$std_resid, na.rm = TRUE)

ggplot(datum_clean, aes(x = std_resid)) +
  # 1. Create the Histogram (using density scale)
  geom_histogram(aes(y = after_stat(density)),
                 bins = 15, # Adjust bin count for best visualization
                 color = "black",
                 fill = "lightblue") +

  # 2. Overlay the Density of the Residuals (Green Line)
  geom_density(color = "darkgreen", linewidth = 1) +

  # 3. Overlay the Theoretical Normal Density Curve (Red Dashed Line)
  stat_function(fun = dnorm,
                args = list(mean = resid_mean, sd = resid_sd),
                color = "red",
                linewidth = 1,
                linetype = "dashed") +

  labs(
    title = "Residual Normality Check: Histogram with Density Curve",
    x = "Standardized Residuals (Z-Scores)",
    y = "Density"
  ) +
  theme_minimal()
```



Conclusion: Overall, the residuals do not look bad. It does look like there is one outlier, so there are many ways to proceed.

Next Steps: Remove the outlier and see how that changes the results.

```
#Removing the outlier
  #Since it only looks like there is one outlier, I decided to run the Grubbs test, whic
grubbs.test(datumlong2$Ketones)
```

```

	Grubbs test for one outlier

data:  datumlong2$Ketones
G = 4.75003, U = 0.84853, p-value = 5.932e-05
alternative hypothesis: highest value 1.2 is an outlier
```

```
  #Making the new data frame
datumlong2 = datumlong2[-149, ]

#Running the models
#  1. Start with global linear mixed effects model to assess ketone levels
results = lmer(Ketones ~ TreatmentF * SexF * TimepointF + (1|ID),
               data = datumlong2, na.action = na.omit)
summary(results)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ TreatmentF * SexF * TimepointF + (1 | ID)
   Data: datumlong2

REML criterion at convergence: -146.8

Scaled residuals: 
     Min       1Q   Median       3Q      Max 
-2.22002 -0.57941 -0.05381  0.57506  2.63632 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.004711 0.06864 
 Residual             0.013502 0.11620 
Number of obs: 150, groups:  ID, 82

Fixed effects:
                                   Estimate Std. Error        df t value
(Intercept)                         0.36865    0.03170 136.07810  11.628
TreatmentFHeat                      0.14608    0.04539 137.90781   3.218
SexFM                               0.16813    0.04487 135.34370   3.747
TimepointFPre                       0.08385    0.03801  63.98577   2.206
TreatmentFHeat:SexFM               -0.13843    0.06424 137.35093  -2.155
TreatmentFHeat:TimepointFPre       -0.14631    0.05384  66.46758  -2.717
SexFM:TimepointFPre                -0.16718    0.05499  65.23723  -3.040
TreatmentFHeat:SexFM:TimepointFPre  0.10664    0.07717  66.70011   1.382
                                   Pr(>|t|)    
(Intercept)                         < 2e-16 ***
TreatmentFHeat                     0.001607 ** 
SexFM                              0.000264 ***
TimepointFPre                      0.030982 *  
TreatmentFHeat:SexFM               0.032906 *  
TreatmentFHeat:TimepointFPre       0.008385 ** 
SexFM:TimepointFPre                0.003401 ** 
TreatmentFHeat:SexFM:TimepointFPre 0.171621    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TrFH:SFM TFH:TF SFM:TF
TreatmntFHt -0.698                                            
SexFM       -0.707  0.493                                     
TimepointFPr -0.639  0.446  0.451                              
TrtmFH:SFM   0.494 -0.707 -0.699 -0.315                        
TrtmFH:TFP   0.451 -0.659 -0.319 -0.706  0.466                 
SxFM:TmpnFP  0.441 -0.308 -0.621 -0.691  0.434    0.488        
TFH:SFM:TFP -0.315  0.460  0.442  0.493 -0.647   -0.698 -0.713 
```

```
Anova(results, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
                              Chisq Df Pr(>Chisq)    
(Intercept)                135.2204  1  < 2.2e-16 ***
TreatmentF                  10.3586  1   0.001289 ** 
SexF                        14.0393  1   0.000179 ***
TimepointF                   4.8664  1   0.027384 *  
TreatmentF:SexF              4.6440  1   0.031163 *  
TreatmentF:TimepointF        7.3834  1   0.006583 ** 
SexF:TimepointF              9.2428  1   0.002364 ** 
TreatmentF:SexF:TimepointF   1.9096  1   0.167010    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 2. The three way interaction was not significant, so it was removed and all two way in
results2 = lmer(Ketones ~ TreatmentF * SexF + TreatmentF * TimepointF +
                SexF * TimepointF + (1|ID), data = datumlong2, na.action = na.omit)
summary(results2)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ TreatmentF * SexF + TreatmentF * TimepointF + SexF *  
    TimepointF + (1 | ID)
   Data: datumlong2

REML criterion at convergence: -148.2

Scaled residuals: 
     Min       1Q   Median       3Q      Max 
-2.09141 -0.57630 -0.07907  0.54594  2.73164 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.004591 0.06775 
 Residual             0.013702 0.11706 
Number of obs: 150, groups:  ID, 82

Fixed effects:
                             Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                   0.38245    0.03015 127.54365  12.685  < 2e-16
TreatmentFHeat                0.11706    0.04036 117.31971   2.900 0.004454
SexFM                         0.14066    0.04029 110.49575   3.491 0.000692
TimepointFPre                 0.05795    0.03331  65.90339   1.739 0.086616
TreatmentFHeat:SexFM         -0.08093    0.04897  70.68324  -1.653 0.102817
TreatmentFHeat:TimepointFPre -0.09420    0.03884  67.78833  -2.425 0.017969
SexFM:TimepointFPre          -0.11309    0.03885  67.84895  -2.911 0.004871
                                
(Intercept)                  ***
TreatmentFHeat               ** 
SexFM                        ***
TimepointFPre                .  
TreatmentFHeat:SexFM            
TreatmentFHeat:TimepointFPre *  
SexFM:TimepointFPre          ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TFH:SF TFH:TF
TreatmntFHt -0.656                                   
SexFM       -0.666  0.363                            
TimepointFPr -0.588  0.286  0.301                     
TrtmFH:SFM   0.400 -0.603 -0.602  0.005              
TrtmFH:TFP   0.342 -0.534 -0.015 -0.581  0.026       
SxFM:TmpnFP  0.328  0.031 -0.488 -0.557 -0.051 -0.018
```

```
Anova(results2, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
                         Chisq Df Pr(>Chisq)    
(Intercept)           160.9091  1  < 2.2e-16 ***
TreatmentF              8.4110  1  0.0037296 ** 
SexF                   12.1886  1  0.0004808 ***
TimepointF              3.0258  1  0.0819477 .  
TreatmentF:SexF         2.7316  1  0.0983820 .  
TreatmentF:TimepointF   5.8816  1  0.0152995 *  
SexF:TimepointF         8.4736  1  0.0036033 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 3. The Treatment*Sex interaction was not significant, so it was removed from the model
results3 = lmer(Ketones ~ SexF * TimepointF + TreatmentF * TimepointF +
                (1|ID), data = datumlong2, na.action = na.omit)
summary(results3)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ SexF * TimepointF + TreatmentF * TimepointF + (1 |      ID)
   Data: datumlong2

REML criterion at convergence: -149.7

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-1.9851 -0.5714 -0.1066  0.5615  2.8285 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.004778 0.06912 
 Residual             0.013765 0.11732 
Number of obs: 150, groups:  ID, 82

Fixed effects:
                             Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                   0.40237    0.02782 138.41317  14.463  < 2e-16
SexFM                         0.10058    0.03239 139.28444   3.105  0.00231
TimepointFPre                 0.05825    0.03340  65.43856   1.744  0.08583
TreatmentFHeat                0.07695    0.03241 139.26888   2.375  0.01894
SexFM:TimepointFPre          -0.11630    0.03890  67.48168  -2.990  0.00389
TimepointFPre:TreatmentFHeat -0.09269    0.03893  67.33604  -2.381  0.02010
                                
(Intercept)                  ***
SexFM                        ** 
TimepointFPre                .  
TreatmentFHeat               *  
SexFM:TimepointFPre          ** 
TimepointFPre:TreatmentFHeat *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) SexFM  TmpnFP TrtmFH SFM:TF
SexFM       -0.581                            
TimepointFPr -0.642  0.379                     
TreatmntFHt -0.568  0.000  0.361              
SxFM:TmpnFP  0.379 -0.648 -0.558  0.000       
TmpntFP:TFH  0.360  0.001 -0.581 -0.648 -0.017
```

```
Anova(results3, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
                         Chisq Df Pr(>Chisq)    
(Intercept)           209.1729  1  < 2.2e-16 ***
SexF                    9.6395  1   0.001904 ** 
TimepointF              3.0420  1   0.081136 .  
TreatmentF              5.6383  1   0.017572 *  
SexF:TimepointF         8.9379  1   0.002793 ** 
TimepointF:TreatmentF   5.6689  1   0.017268 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Use AIC to test which one of these models has the best fit
AIC(results, results2, results3)
```

```
         df       AIC
results  10 -126.8394
results2  9 -130.2221
results3  8 -133.6991
```

```
#This is the global model, use 'results3' for testing the assumptions

results3 = lmer(Ketones ~ SexF * TimepointF + TreatmentF * TimepointF +
                (1|ID), data = datumlong2, na.action = na.omit)
summary(results3)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ SexF * TimepointF + TreatmentF * TimepointF + (1 |      ID)
   Data: datumlong2

REML criterion at convergence: -149.7

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-1.9851 -0.5714 -0.1066  0.5615  2.8285 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.004778 0.06912 
 Residual             0.013765 0.11732 
Number of obs: 150, groups:  ID, 82

Fixed effects:
                             Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                   0.40237    0.02782 138.41317  14.463  < 2e-16
SexFM                         0.10058    0.03239 139.28444   3.105  0.00231
TimepointFPre                 0.05825    0.03340  65.43856   1.744  0.08583
TreatmentFHeat                0.07695    0.03241 139.26888   2.375  0.01894
SexFM:TimepointFPre          -0.11630    0.03890  67.48168  -2.990  0.00389
TimepointFPre:TreatmentFHeat -0.09269    0.03893  67.33604  -2.381  0.02010
                                
(Intercept)                  ***
SexFM                        ** 
TimepointFPre                .  
TreatmentFHeat               *  
SexFM:TimepointFPre          ** 
TimepointFPre:TreatmentFHeat *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) SexFM  TmpnFP TrtmFH SFM:TF
SexFM       -0.581                            
TimepointFPr -0.642  0.379                     
TreatmntFHt -0.568  0.000  0.361              
SxFM:TmpnFP  0.379 -0.648 -0.558  0.000       
TmpntFP:TFH  0.360  0.001 -0.581 -0.648 -0.017
```

```
Anova(results3, type = "III")
```

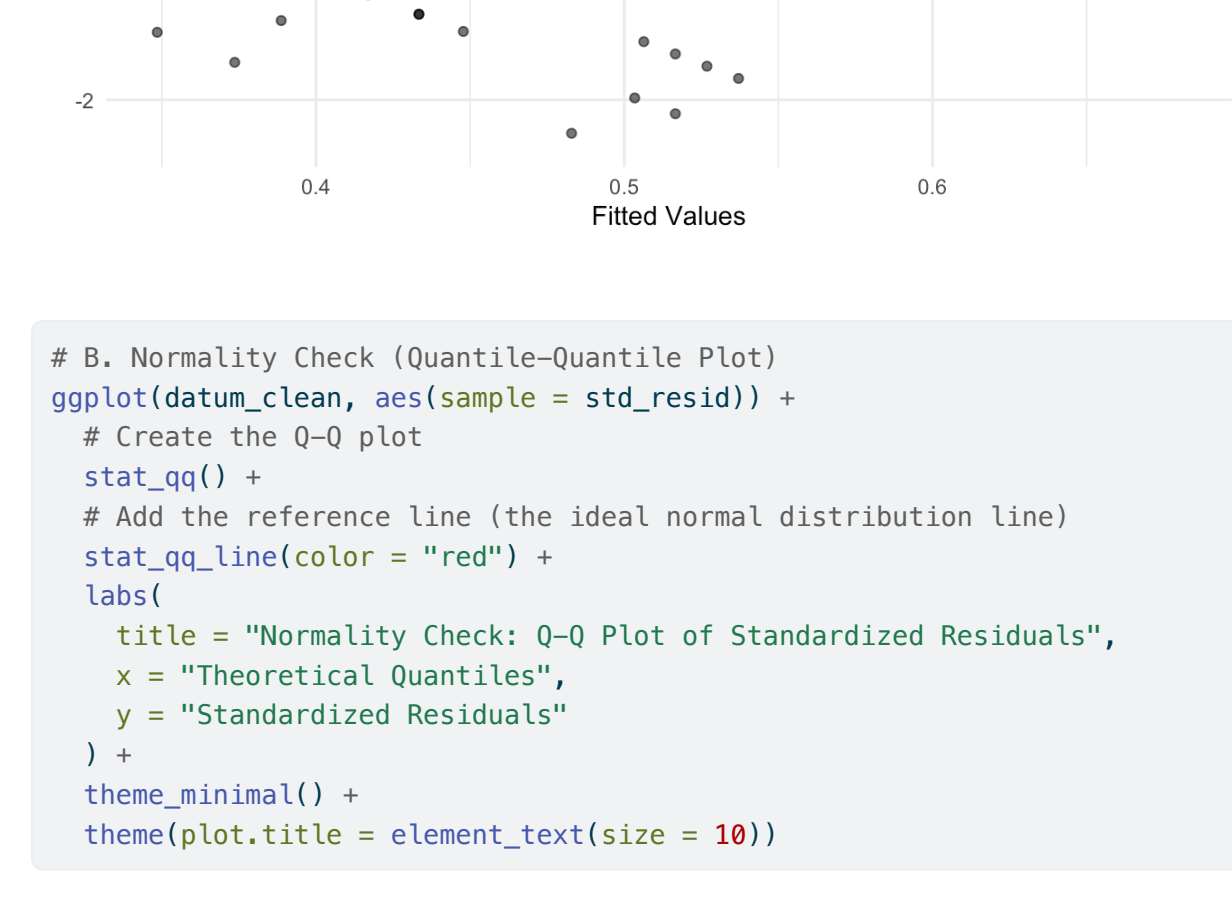
```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
                         Chisq Df Pr(>Chisq)    
(Intercept)           209.1729  1  < 2.2e-16 ***
SexF                    9.6395  1   0.001904 ** 
TimepointF              3.0420  1   0.081136 .  
TreatmentF              5.6383  1   0.017572 *  
SexF:TimepointF         8.9379  1   0.002793 ** 
TimepointF:TreatmentF   5.6689  1   0.017268 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Assumptions of the model


# 1. Check the VIF of this model
vif(results3)
```

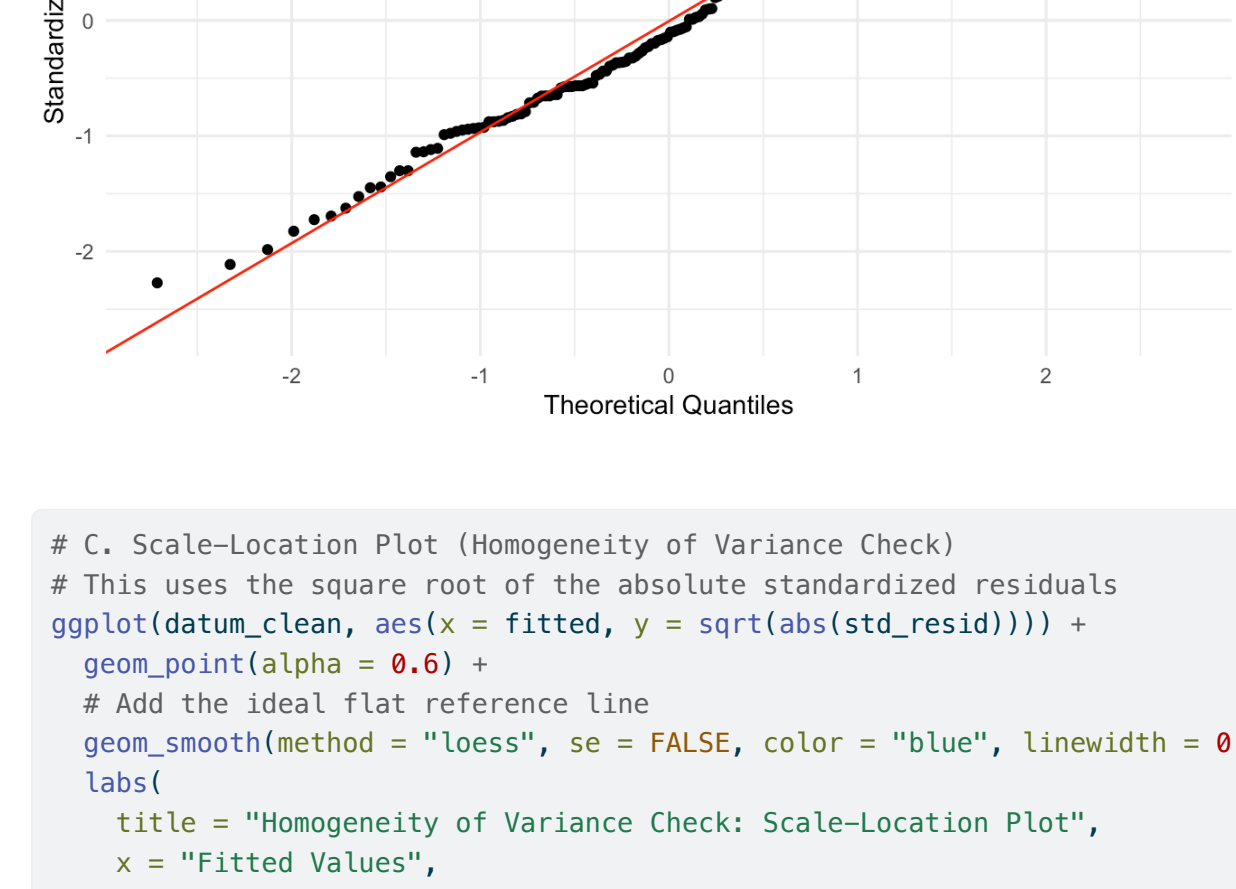
```
                 SexF            TimepointF            TreatmentF 
             1.726785              2.953220              1.727576 
      SexF:TimepointF TimepointF:TreatmentF 
             2.640558              2.843044 
```

```
  #VIF = 1: No multicollinearity
  #VIF between 1 and 5: Generally considered acceptable or moderate multicollinearity
  #VIF > 5 (or > 10): Indicates potentially high multicollinearity

# 2. Check other assumptions of a linear model

# --- Extract Residuals and Fitted Values ---
#1. Extract the data frame used by the model
datum_clean <- model.frame(results3)

# 2. Calculate and assign the fitted and residual values to this clean data frame
datum_clean$fitted <- fitted(results3)
datum_clean$raw_resid <- residuals(results3)
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)

# 3. Use the studentized residuals method from the 'lme4' utility functions
# Although 'rstandard' doesn't work, 'residuals' is robust.
# If you really need a standardized residual, use this function:
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)


# --- 2. Create the Plots ---

# A. Linearity and Homoscedasticity Plot (Residuals vs. Fitted)
ggplot(datum_clean, aes(x = fitted, y = std_resid)) +
  geom_point(alpha = 0.6) +
  # Add the ideal reference line at y=0
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  # Add a LOESS smoothing line (red line in performance)
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Linearity Check: Standardized Residuals vs. Fitted Values",
    x = "Fitted Values",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

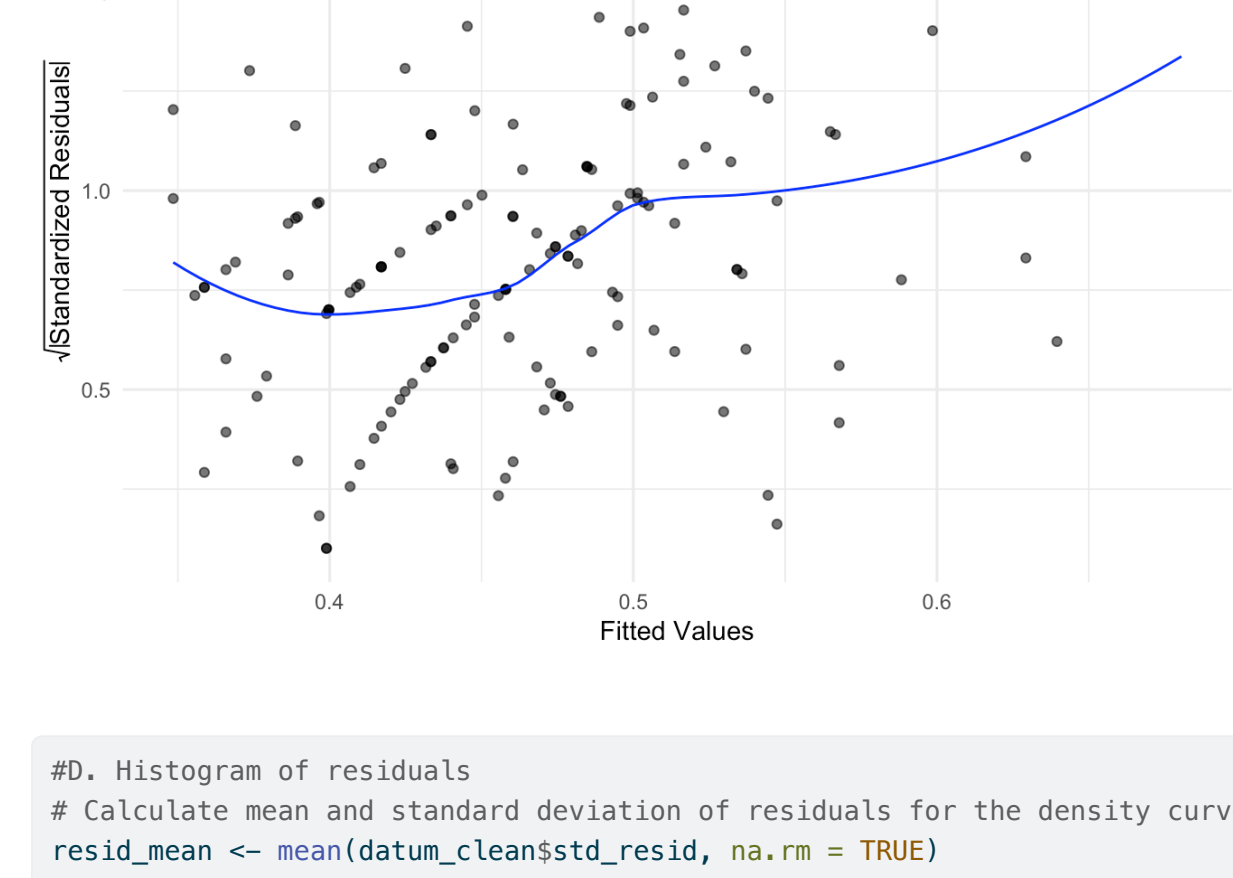


```
# B. Normality Check (Quantile-Quantile Plot)
ggplot(datum_clean, aes(sample = std_resid)) +
  # Create the Q-Q plot
  stat_qq() +
  # Add the reference line (the ideal normal distribution line)
  stat_qq_line(color = "red") +
  labs(
    title = "Normality Check: Q-Q Plot of Standardized Residuals",
    x = "Theoretical Quantiles",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

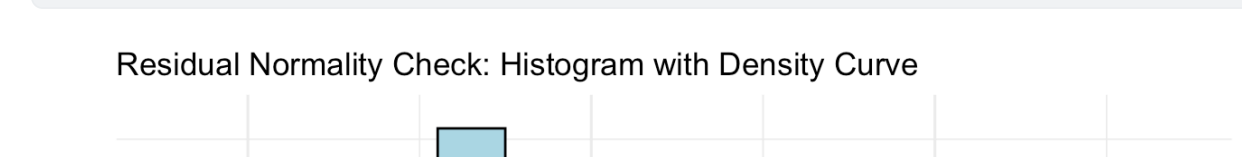


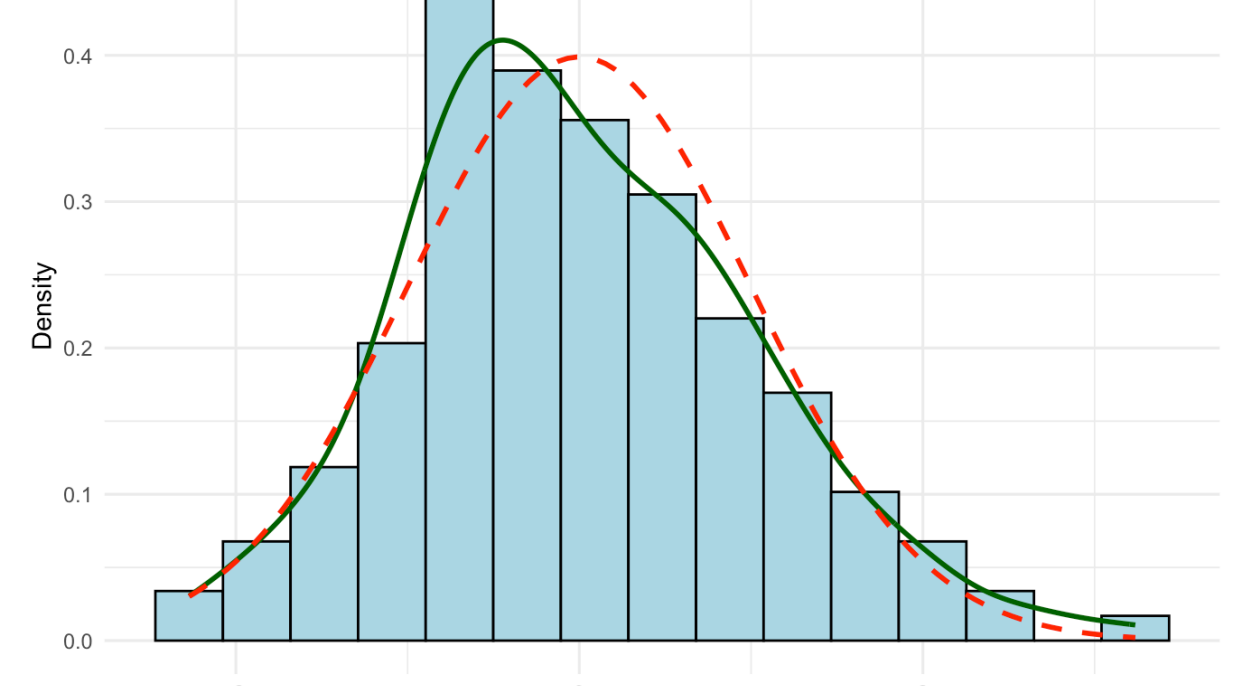
```
# C. Scale-Location Plot (Homogeneity of Variance Check)
# This uses the square root of the absolute standardized residuals
ggplot(datum_clean, aes(x = fitted, y = sqrt(abs(std_resid)))) +
  geom_point(alpha = 0.6) +
  # Add the ideal flat reference line
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Homogeneity of Variance Check: Scale-Location Plot",
    x = "Fitted Values",
    y = expression(sqrt(abs("Standardized Residuals")))
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
#D. Histogram of residuals
# Calculate mean and standard deviation of residuals for the density curve
resid_mean <- mean(datum_clean$std_resid, na.rm = TRUE)
resid_sd <- sd(datum_clean$std_resid, na.rm = TRUE)

ggplot(datum_clean, aes(x = std_resid)) +
  # 1. Create the Histogram (using density scale)
  geom_histogram(aes(y = after_stat(density)),
                 bins = 15, # Adjust bin count for best visualization
                 color = "black",
                 fill = "lightblue") +

  # 2. Overlay the Density of the Residuals (Green Line)
  geom_density(color = "darkgreen", linewidth = 1) +

  # 3. Overlay the Theoretical Normal Density Curve (Red Dashed Line)
  stat_function(fun = dnorm,
                args = list(mean = resid_mean, sd = resid_sd),
                color = "red",
                linewidth = 1,
                linetype = "dashed") +

  labs(
    title = "Residual Normality Check: Histogram with Density Curve",
    x = "Standardized Residuals (Z-Scores)",
    y = "Density"
  ) +
  theme_minimal()
```





Conclusion: Residuals look better after removal of the outlier. However, the stats do not really change. Both interactions have smaller p-values after removing the outlier, so the outlier is not driving significance.

Next Steps: Keep the outlier in and report those results. Also, so those results for post hoc. Can say we ran it both ways and the interactions had smaller p-values after outlier removal.

```
# 1. Since there are two 2-way interactions, we will subset all males to look at time po
datumlongMale = datumlong2 %>%
  filter(SexF == "M")

resultsmale = lmer(Ketones ~ TimepointF + (1|ID),
               data = datumlongMale, na.action = na.omit)
summary(resultsmale)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ TimepointF + (1 | ID)
   Data: datumlongMale

REML criterion at convergence: -62.1

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.99126 -0.65396 -0.02522  0.46758  2.51057

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.003669 0.06057
 Residual             0.018661 0.13660
Number of obs: 73, groups:  ID, 40

Fixed effects:
              Estimate Std. Error       df t value Pr(>|t|)
(Intercept)    0.53962    0.02522 69.83476  21.400  < 2e-16 ***
TimepointFPre -0.10446    0.03226 32.14449  -3.238  0.00279 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr)
TimepontFPr -0.670
```

```
Anova(resultsmale, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
              Chisq Df Pr(>Chisq)
(Intercept) 457.963  1  < 2.2e-16 ***
TimepointF   10.486  1   0.001203 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 2. Since there are two 2-way interactions, we will subset all females to look at time
datumlongFemale = datumlong2 %>%
  filter(SexF == "F")
resultsfemale = lmer(Ketones ~ TimepointF + (1|ID),
               data = datumlongFemale, na.action = na.omit)
summary(resultsfemale)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ TimepointF + (1 | ID)
   Data: datumlongFemale

REML criterion at convergence: -93.8

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.6816 -0.5704 -0.2117  0.7923  2.7632

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.003697 0.0608
 Residual             0.011912 0.1091
Number of obs: 77, groups:  ID, 42

Fixed effects:
              Estimate Std. Error       df t value Pr(>|t|)
(Intercept)    0.43940    0.02102 73.47215  20.905   <2e-16 ***
TimepointFPre  0.01298    0.02525 38.65444   0.514     0.61
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr)
TimepontFPr -0.667
```

```
Anova(resultsfemale, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
               Chisq Df Pr(>Chisq)
(Intercept) 437.0270  1     <2e-16 ***
TimepointF    0.2644  1     0.6071
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 3. Since there are two 2-way interactions, we will subset all treatments to look at ti
datumlongHeat = datumlong2 %>%
  filter(TreatmentF == "Heat")

resultsheat = lmer(Ketones ~ TimepointF + (1|ID),
               data = datumlongHeat, na.action = na.omit)
summary(resultsheat)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ TimepointF + (1 | ID)
   Data: datumlongHeat

REML criterion at convergence: -74

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.40176 -0.62453 -0.06476  0.61356  2.46409

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.006263 0.07914
 Residual             0.014402 0.12001
Number of obs: 77, groups:  ID, 43

Fixed effects:
              Estimate Std. Error       df t value Pr(>|t|)
(Intercept)    0.53048    0.02442 73.08384  21.727  < 2e-16 ***
TimepointFPre -0.09327    0.02802 37.00545  -3.328  0.00199 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr)
TimepontFPr -0.658
```

```
Anova(resultsheat, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
              Chisq Df Pr(>Chisq)
(Intercept) 472.082  1  < 2.2e-16 ***
TimepointF   11.077  1  0.0008739 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 2. Since there are two 2-way interactions, we will subset all females to look at time
datumlongControl = datumlong2 %>%
  filter(TreatmentF == "Control")
resultscontrol = lmer(Ketones ~ TimepointF + (1|ID),
               data = datumlongControl, na.action = na.omit)
summary(resultscontrol)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Ketones ~ TimepointF + (1 | ID)
   Data: datumlongControl

REML criterion at convergence: -75.7

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.7534 -0.6171 -0.1946  0.5841  3.4872

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.001833 0.04281
 Residual             0.016462 0.12830
Number of obs: 73, groups:  ID, 39

Fixed effects:
                Estimate Std. Error         df t value Pr(>|t|)
(Intercept)    0.4527576  0.0225343 70.3547006  20.092   <2e-16 ***
TimepointFPre -0.0003379  0.0301397 30.4669017  -0.011    0.991
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr)
TimepontFPr -0.679
```

```
Anova(resultscontrol, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Ketones
               Chisq Df Pr(>Chisq)
(Intercept) 403.6866  1     <2e-16 ***
TimepointF    0.0001  1     0.9911
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Post Hoc Testing

  #Test 1: Compare Timepoints (Pre vs. Post) separately for each Sex
posthoc_Sex_Time = emmeans(results_final, ~ TimepointF * SexF)
posthoc_Sex_Time1 = pairs(posthoc_Sex_Time, adjust = "sidak")
print(posthoc_Sex_Time1)
```

```
 contrast        estimate     SE    df t.ratio p.value
 Post F - Pre F   0.00771 0.0305  73.3   0.253  0.9999
 Post F - Post M -0.08081 0.0351 142.6  -2.304  0.1286
 Post F - Pre M   0.02386 0.0344 142.1   0.693  0.9822
 Pre F - Post M  -0.08851 0.0338 141.6  -2.616  0.0577
 Pre F - Pre M    0.01615 0.0331 140.8   0.488  0.9973
 Post M - Pre M   0.10466 0.0315  74.9   3.320  0.0083

Results are averaged over the levels of: TreatmentF
Degrees-of-freedom method: kenward-roger
P value adjustment: sidak method for 6 tests
```

```
Sex_Time_output = cld(posthoc_Sex_Time,
                  adjust = "sidak",
                  Letters = letters)

print(Sex_Time_output)
```

```
  TimepointF SexF    emmean         SE       df  lower.CL  upper.CL .group
4        Pre    M 0.4365958 0.02403782 141.3092 0.3759472 0.4972445     a
2        Pre    F 0.4527452 0.02282307 140.3376 0.3951563 0.5103341     ab
1       Post    F 0.4604532 0.02462198 142.6883 0.3983384 0.5225680     ab
3       Post    M 0.5412593 0.02497824 142.4311 0.4782444 0.6042743      b
```

```
  #Test 2: Compare Timepoints (Pre vs. Post) separately for each Treatment
posthoc_Treat_Time = emmeans(results_final, ~ TimepointF * TreatmentF)
posthoc_Treat_Time1 = pairs(posthoc_Treat_Time, adjust = "sidak")
print(posthoc_Treat_Time1)
```

```
 contrast                    estimate     SE    df t.ratio p.value
 Post Control - Pre Control  0.000378 0.0314  72.9   0.012  1.0000
 Post Control - Post Heat   -0.096317 0.0351 142.6  -2.746  0.0402
 Post Control - Pre Heat     0.015676 0.0334 141.1   0.470  0.9978
 Pre Control - Post Heat    -0.096695 0.0349 142.4  -2.773  0.0372
 Pre Control - Pre Heat      0.015298 0.0332 140.9   0.461  0.9980
 Post Heat - Pre Heat        0.111993 0.0306  75.2   3.658  0.0028

Results are averaged over the levels of: SexF
Degrees-of-freedom method: kenward-roger
P value adjustment: sidak method for 6 tests
```

```
Treat_Time_output = cld(posthoc_Treat_Time,
                  adjust = "sidak",
                  Letters = letters)

print(Treat_Time_output)
```

```
  TimepointF TreatmentF    emmean         SE       df  lower.CL  upper.CL
4        Pre       Heat 0.4370215 0.02254592 140.3371 0.3801319 0.4939111
2        Pre    Control 0.4523195 0.02433325 141.3096 0.3909255 0.5137135
1       Post    Control 0.4526979 0.02462892 141.6947 0.3905600 0.5148357
3       Post       Heat 0.5490147 0.02497151 143.2670 0.4860214 0.6120080
  .group
4     a
2     a
1     a
3      b
```

What does this means?

For the Timepoint*Sex interaction

- Males at the ‘post’ time point had higher ketone levels than males at the ‘pre’ time point
- Weirdly, ‘pre’ and ‘post’ females did not statistically differ from ‘pre’ or ‘post’ males

For the Timepoint*Treatment interaction

- Heated individuals had higher ketone levels at the ‘post’ time point compared to the ‘pre’ time point
- The ‘post’ heated time point also had higher ketone levels compared to controls at both ‘pre’ and ‘post’ timepoints

Analysis - Body Temperature and Ketone Correlation

```
#Removing the outlier
  #I ran the test with the outlier included and the assumptions were not met, so I decid
  #Since it only looks like there is one outlier, I decided to run the Grubbs test, whic
grubbs.test(datum2$PostKetones)
```

```

    Grubbs test for one outlier

data:  datum2$PostKetones
G = 3.81689, U = 0.70874, p-value = 0.0009481
alternative hypothesis: highest value 1.2 is an outlier
```

```
  #Making the new data frame
datum3 = datum2[-51, ]


#Analysis

# 1. Run Pearson's Correlation Test for both heated and control groups
pearson_result = cor.test(
  x = datum2$PostKetones,
  y = datum2$AvgMaxTemp,
  method = "pearson"
)
  # View the results
print(pearson_result)
```

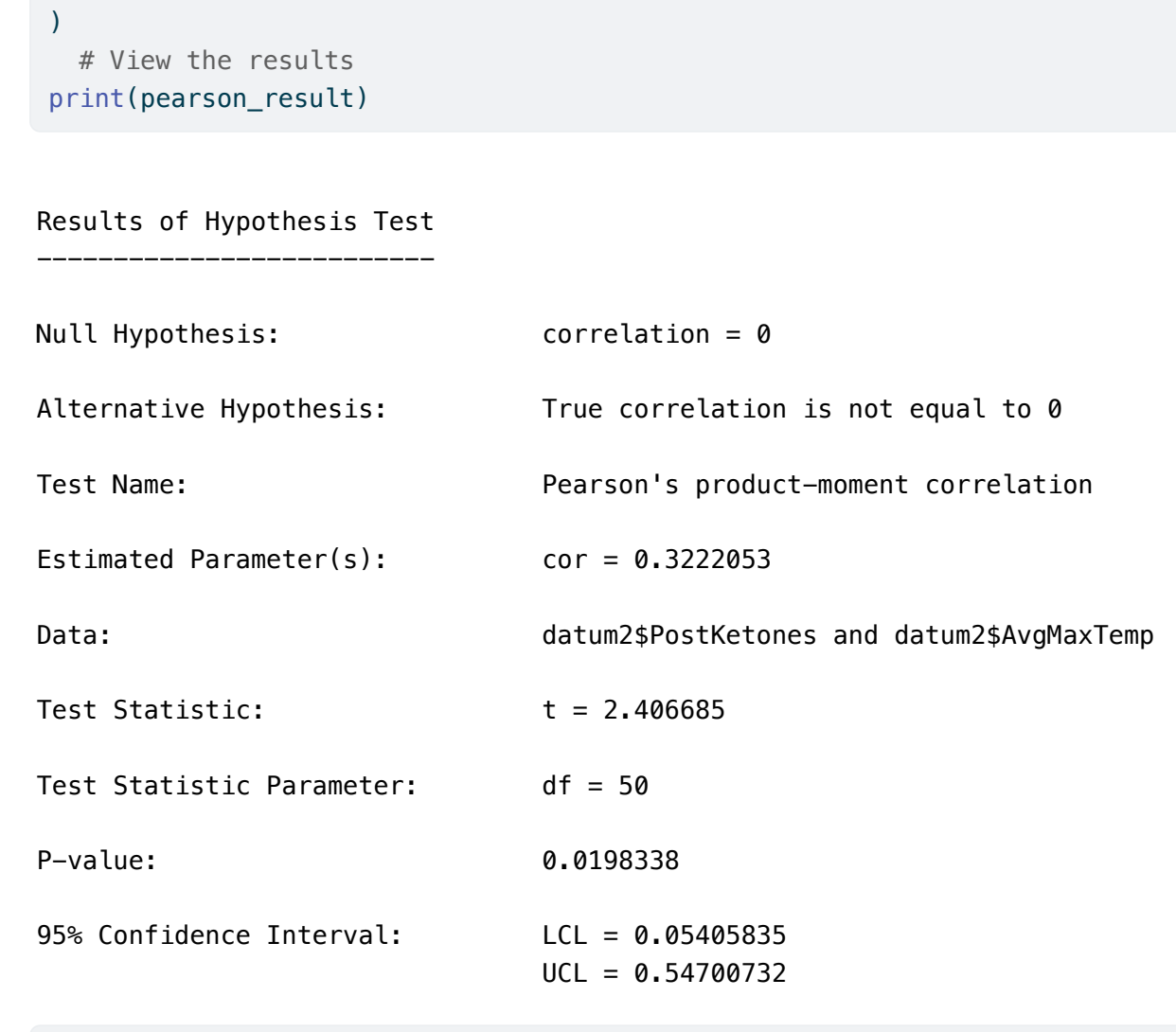
```

Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = 0.3222053

Data:                            datum2$PostKetones and datum2$AvgMaxTemp

Test Statistic:                  t = 2.406685

Test Statistic Parameter:        df = 50

P-value:                         0.0198338

95% Confidence Interval:         LCL = 0.05405835
                                 UCL = 0.54700732
```

```
# 2.  Run Pearson's Correlation Test heated group
datumHeated = datum2 %>%
 filter(TreatmentF == "Heat")

pearson_resultHeat = cor.test(
  x = datumHeated$PostKetones,
  y = datumHeated$AvgMaxTemp,
  method = "pearson"
)
  # View the results
print(pearson_resultHeat)
```

```

Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = 0.3508277

Data:                            datumHeated$PostKetones and datumHeated$AvgMaxTemp

Test Statistic:                  t = 1.873199

Test Statistic Parameter:        df = 25

P-value:                         0.07277884

95% Confidence Interval:         LCL = -0.0336759
                                 UCL =  0.6448682
```

```
ggplot(data=datumHeated, aes(AvgFinalTemp, PostKetones)) +
  geom_point()
```

```
Warning: Removed 12 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```
# 2.  Run Pearson's Correlation Test control group
datumControl = datum2 %>%
 filter(TreatmentF == "Control")

pearson_resultControl = cor.test(
  x = datumControl$PostKetones,
  y = datumControl$AvgMaxTemp,
  method = "pearson"
)
  # View the results
print(pearson_resultControl)
```

```

Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = 0.1124532

Data:                            datumControl$PostKetones and datumControl$AvgMaxTemp

Test Statistic:                  t = 0.5427491

Test Statistic Parameter:        df = 23

P-value:                         0.5925237

95% Confidence Interval:         LCL = -0.2958222
                                 UCL =  0.4859897
```

Manuscript Graph

Graph depicting treatments impact on ketones

```
# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(TreatmentF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_Ketones = mean(Ketones, na.rm = TRUE),
    SD_Ketones = sd(Ketones, na.rm = TRUE),
    SE_Ketones = SD_Ketones / sqrt(N_group),
    CI_Margin_Ketones = 1.96 * SE_Ketones,
    CI_Upper_Group = Mean_Ketones + CI_Margin_Ketones,
    CI_Lower_Group = Mean_Ketones - CI_Margin_Ketones
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 35
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl>   <dbl>
1  4818 M     Control   Poseiden  Right Pre       No    3/3/25   18.0    237
2  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6    198.
3  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1    192.
4  4822 M     Heat      Poseiden  Right Pre       No    3/6/25   16.4    204
5  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4    239
6  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5    226.
# ℹ 25 more variables: Ketones <dbl>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, N_group <int>, Mean_Ketones <dbl>, SD_Ketones <dbl>,
#   SE_Ketones <dbl>, CI_Margin_Ketones <dbl>, CI_Upper_Group <dbl>,
#   CI_Lower_Group <dbl>
```

```
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.4
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$TreatmentF = as.factor(plot_data$TreatmentF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = Ketones, color = TreatmentF)) +
  geom_jitter(
    size = 3.5,
    alpha = 0.5,
    position = position_jitterdodge(jitter.width = 0.2, dodge.width = DODGE_WIDTH)
  ) +

  geom_point(
    aes(y = Mean_Ketones, group = TreatmentF),
    size = 7.5,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = TreatmentF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_Ketones, group = TreatmentF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "Control" = "#AAB8C2",
      "Heat" = "firebrick4"
    )
  ) +

  labs(
    x = "Time Point",
    y = "Ketone (mmol/L)",
    color = "Treatment Group"
  ) +

  ylim(0,1.4) +

  theme_classic() +

    theme(
    axis.title = element_text(face = "bold", size = 30),
    axis.text = element_text(size = 25),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 28),
    legend.text = element_text(size = 25),
    strip.text = element_text(size = 20),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA),
    legend.box.background = element_rect(fill = "transparent", color = NA),
    axis.line = element_line(linewidth = 1.5, colour = "black")
  )



# 4. Adding signifigance and sample size
plot1 =
plot +
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    y = c(.7 + .2
          , .7 + .2, .95 + .2, 1.2 + .2),
    label = c("a", "a", "a", "b"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    # Adjust the y-coordinates to be slightly lower
     y = c(.7 + .1, .7 + .1, .95 + .1, 1.2 + .1),
    # Add your sample sizes here
    label = c("(37)", "(43)", "(35)", "(34)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")

# 5. Save the graph
ggsave(
  filename = "KetonePlot1.png",
  plot = plot1,
  width = 11.04,
  height = 6.49,
  units = "in",
  dpi = 300,
  bg = "transparent"
)
```

Manuscript Graph - With Sex

```
# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(TreatmentF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_Ketones = mean(Ketones, na.rm = TRUE),
    SD_Ketones = sd(Ketones, na.rm = TRUE),
    SE_Ketones = SD_Ketones / sqrt(N_group),
    CI_Margin_Ketones = 1.96 * SE_Ketones,
    CI_Upper_Group = Mean_Ketones + CI_Margin_Ketones,
    CI_Lower_Group = Mean_Ketones - CI_Margin_Ketones
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 35
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl>   <dbl>
1  4818 M     Control   Poseiden  Right Pre       No    3/3/25   18.0    237
2  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6    198.
3  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1    192.
4  4822 M     Heat      Poseiden  Right Pre       No    3/6/25   16.4    204
5  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4    239
6  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5    226.
# ℹ 25 more variables: Ketones <dbl>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, N_group <int>, Mean_Ketones <dbl>, SD_Ketones <dbl>,
#   SE_Ketones <dbl>, CI_Margin_Ketones <dbl>, CI_Upper_Group <dbl>,
#   CI_Lower_Group <dbl>
```

```
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.4
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$TreatmentF = as.factor(plot_data$TreatmentF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = Ketones, color = TreatmentF)) +

  geom_jitter(
    aes(shape = SexF, group = TreatmentF),
    size = 3.5,
    alpha = 0.5,
    position = position_jitterdodge(
    jitter.width = 0.2,
    dodge.width = DODGE_WIDTH
  )
) +

  geom_point(
    aes(y = Mean_Ketones, group = TreatmentF),
    size = 7.5,
    shape = 18,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = TreatmentF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_Ketones, group = TreatmentF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "Control" = "#AAB8C2",
      "Heat" = "firebrick4"
    )
  ) +

  labs(
    x = "Time Point",
    y = "Ketone (mmol/L)",
    color = "Treatment Group"
  ) +

  guides(shape = "none") +

  ylim(0,1.4) +

  theme_classic() +

    theme(
    axis.title = element_text(face = "bold", size = 30),
    axis.text = element_text(size = 25),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 28),
    legend.text = element_text(size = 25),
    strip.text = element_text(size = 20),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA),
    legend.box.background = element_rect(fill = "transparent", color = NA),
    axis.line = element_line(linewidth = 1.5, colour = "black")
  )



# 4. Adding signifigance and sample size
plot1 =
plot +
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    y = c(.7 + .2
          , .7 + .2, .95 + .2, 1.2 + .2),
    label = c("a", "a", "a", "b"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    # Adjust the y-coordinates to be slightly lower
     y = c(.7 + .1, .7 + .1, .95 + .1, 1.2 + .1),
    # Add your sample sizes here
    label = c("(37)", "(43)", "(35)", "(34)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")

# 5. Save the graph
ggsave(
  filename = "KetonePlotSex.png",
  plot = plot1,
  width = 11.04,
  height = 6.49,
  units = "in",
  dpi = 300,
  bg = "transparent"
)
```

Manuscript Graph - Sex and time point interaction

```
# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(SexF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_Ketones = mean(Ketones, na.rm = TRUE),
    SD_Ketones = sd(Ketones, na.rm = TRUE),
    SE_Ketones = SD_Ketones / sqrt(N_group),
    CI_Margin_Ketones = 1.96 * SE_Ketones,
    CI_Upper_Group = Mean_Ketones + CI_Margin_Ketones,
    CI_Lower_Group = Mean_Ketones - CI_Margin_Ketones
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 35
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl>   <dbl>
1  4818 M     Control   Poseiden  Right Pre       No    3/3/25   18.0    237
2  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6    198.
3  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1    192.
4  4822 M     Heat      Poseiden  Right Pre       No    3/6/25   16.4    204
5  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4    239
6  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5    226.
# ℹ 25 more variables: Ketones <dbl>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, N_group <int>, Mean_Ketones <dbl>, SD_Ketones <dbl>,
#   SE_Ketones <dbl>, CI_Margin_Ketones <dbl>, CI_Upper_Group <dbl>,
#   CI_Lower_Group <dbl>
```

```
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.4
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$SexF = as.factor(plot_data$SexF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = Ketones, color = SexF)) +
  geom_jitter(
    size = 3.5,
    alpha = 0.5,
    position = position_jitterdodge(jitter.width = 0.2, dodge.width = DODGE_WIDTH)
  ) +

  geom_point(
    aes(y = Mean_Ketones, group = SexF),
    size = 7.5,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = SexF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_Ketones, group = SexF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "F" = "grey82",
      "M" = "black"
    )
  ) +

  labs(
    x = "Time Point",
    y = "Ketone (mmol/L)",
    color = "Sex"
  ) +
  theme_classic() +

  theme(
    axis.title = element_text(face = "bold", size = 25),
    axis.text = element_text(size = 22),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 22),
    legend.text = element_text(size = 20),
    strip.text = element_text(size = 20),
  )


# 4. Adding signifigance and sample size
plot1 =
plot +
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    y = c(.7 + .2, .7 + .2, .8 + .2, .95 + .2),
    label = c("ab", "a", "ab", "b"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    # Adjust the y-coordinates to be slightly lower
     y = c(.7 + .1, .7 + .1, .8 + .1, .95 + .1),
    # Add your sample sizes here
    label = c("(42)", "(38)", "(34)", "(35)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")

# 5. Save the graph
ggsave(
  filename = "KetonePlot2.png",
  plot = plot1,
  width = 11.04,
  height = 6.49,
  units = "in",
  dpi = 300
)
```

Manuscript Graph - All Metabolites Combined

```
# Glucose Graph

# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(TreatmentF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_Glucose_Group = mean(Glucose, na.rm = TRUE),
    SD_Glucose_Group = sd(Glucose, na.rm = TRUE),
    SE_Glucose_Group = SD_Glucose_Group / sqrt(N_group),
    CI_Margin_Glucose = 1.96 * SE_Glucose_Group,
    CI_Upper_Group = Mean_Glucose_Group + CI_Margin_Glucose,
    CI_Lower_Group = Mean_Glucose_Group - CI_Margin_Glucose
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 35
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl>   <dbl>
1  4818 M     Control   Poseiden  Right Pre       No    3/3/25   18.0    237
2  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6    198.
3  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1    192.
4  4822 M     Heat      Poseiden  Right Pre       No    3/6/25   16.4    204
5  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4    239
6  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5    226.
# ℹ 25 more variables: Ketones <dbl>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, N_group <int>, Mean_Glucose_Group <dbl>,
#   SD_Glucose_Group <dbl>, SE_Glucose_Group <dbl>, CI_Margin_Glucose <dbl>,
#   CI_Upper_Group <dbl>, CI_Lower_Group <dbl>
```

```
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.4
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$TreatmentF = as.factor(plot_data$TreatmentF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = Glucose, color = TreatmentF)) +
  geom_jitter(
    size = 3.5,
    alpha = 0.5,
    position = position_jitterdodge(jitter.width = 0.2, dodge.width = DODGE_WIDTH)
  ) +

  geom_point(
    aes(y = Mean_Glucose_Group, group = TreatmentF),
    size = 7.5,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = TreatmentF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_Glucose_Group, group = TreatmentF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "Control" = "#AAB8C2",
      "Heat" = "firebrick4"
    )
  ) +

  labs(
    x = "Time Point",
    y = "Blood Glucose (mg/dL)",
    color = "Treatment Group"
  ) +
  theme_classic() +

  theme(
    axis.title = element_text(face = "bold", size = 30),
    axis.text = element_text(size = 25),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 28),
    legend.text = element_text(size = 25),
    strip.text = element_text(size = 20),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA),
    legend.box.background = element_rect(fill = "transparent", color = NA),
    axis.line = element_line(linewidth = 1.5, colour = "black")
  )


# 4. Adding signifigance and sample size

plot1 =  plot +
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    y = c(379, 467, 370, 514),
    label = c("a", "a", "a", "b"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    # Adjust the y-coordinates to be slightly lower
    y = c(379 - 35, 467 - 35, 370 - 35, 514 - 35),
    # Add your sample sizes here
    label = c("(39)", "(36)", "(44)", "(36)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")



#Ketone Graph Sex by Time point

# 1. Double-check your summary table math first
plot3_summary = plot_data %>%
  group_by(SexF, TimepointF) %>%
  summarise(
    Mean_Ketones = mean(Ketones, na.rm = TRUE),
    SD_Ketones = sd(Ketones, na.rm = TRUE),
    N = n(),
    SE = SD_Ketones / sqrt(N),
    CI_Lower = Mean_Ketones - (1.96 * SE),
    CI_Upper = Mean_Ketones + (1.96 * SE),
    .groups = "drop"
  )
DODGE_WIDTH = 0.6

plot3 = ggplot() +
  # Individual data points
  geom_jitter(
    data = plot_data,
    aes(x = TimepointF, y = Ketones, color = SexF, group = SexF), # Map color to SexF
    shape = 21,
    fill = "white", # Set fill to white (NOT mapped, so it stays white)
    size = 3.5,
    stroke = 0.8,
    alpha = 0.4,
    position = position_jitterdodge(jitter.width = 0.2, dodge.width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    data = plot3_summary,
    aes(x = TimepointF, ymin = CI_Lower, ymax = CI_Upper, group = SexF, color = SexF), #
    width = 0.15,
    linewidth = 1,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_point(
    data = plot3_summary,
    aes(x = TimepointF, y = Mean_Ketones, group = SexF, color = SexF), # Map color here
    shape = 21,
    fill = "white",
    size = 7.5,
    stroke = 2,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    data = plot3_summary,
    aes(x = TimepointF, y = Mean_Ketones, group = SexF, color = SexF), # Map color here
    linewidth = 1.2,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  # THIS PART BRINGS THE COLOR BACK
  scale_color_manual(
    values = c(
      "F" = "goldenrod4",  # Or your chosen color for Female
      "M" = "black"    # Or your chosen color for Male
    )
  ) +

  # Ensure the legend title matches
    labs(
    x = "Time Point",
    y = "Ketone (mmol/L)",
    color = "Sex"
  )


# 4. Adding signifigance and sample size
plot3 =
plot3 +
  annotate(
    "text",
    x = c(.85, 1.15, 1.85, 2.15),
    y = c(.7 + .2, .7 + .2, .8 + .2, .95 + .2),
    label = c("ab", "a", "ab", "b"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.85, 1.15, 1.85, 2.15),
    # Adjust the y-coordinates to be slightly lower
     y = c(.7 + .1, .7 + .1, .8 + .1, .95 + .1),
    # Add your sample sizes here
    label = c("(42)", "(38)", "(34)", "(35)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")


# Ketone Graph Treatment by Time point


# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(TreatmentF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_Ketones = mean(Ketones, na.rm = TRUE),
    SD_Ketones = sd(Ketones, na.rm = TRUE),
    SE_Ketones = SD_Ketones / sqrt(N_group),
    CI_Margin_Ketones = 1.96 * SE_Ketones,
    CI_Upper_Group = Mean_Ketones + CI_Margin_Ketones,
    CI_Lower_Group = Mean_Ketones - CI_Margin_Ketones
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 35
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl>   <dbl>
1  4818 M     Control   Poseiden  Right Pre       No    3/3/25   18.0    237
2  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6    198.
3  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1    192.
4  4822 M     Heat      Poseiden  Right Pre       No    3/6/25   16.4    204
5  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4    239
6  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5    226.
# ℹ 25 more variables: Ketones <dbl>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, N_group <int>, Mean_Ketones <dbl>, SD_Ketones <dbl>,
#   SE_Ketones <dbl>, CI_Margin_Ketones <dbl>, CI_Upper_Group <dbl>,
#   CI_Lower_Group <dbl>
```

```
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.6
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$TreatmentF = as.factor(plot_data$TreatmentF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = Ketones, color = TreatmentF)) +
  geom_jitter(
    size = 3.5,
    alpha = 0.5,
    position = position_jitterdodge(jitter.width = 0.2, dodge.width = DODGE_WIDTH)
  ) +

  geom_point(
    aes(y = Mean_Ketones, group = TreatmentF),
    size = 7.5,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = TreatmentF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_Ketones, group = TreatmentF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "Control" = "#AAB8C2",
      "Heat" = "firebrick4"
    )
  ) +

  labs(
    x = "Time Point",
    y = "Ketone (mmol/L)",
    color = "Treatment Group"
  ) +

  theme_classic() +

    theme(
    axis.title = element_text(face = "bold", size = 30),
    axis.text = element_text(size = 25),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 28),
    legend.text = element_text(size = 25),
    strip.text = element_text(size = 20),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA),
    legend.box.background = element_rect(fill = "transparent", color = NA),
    axis.line = element_line(linewidth = 1.5, colour = "black")
  )



# 4. Adding signifigance and sample size
plot2 =
plot +
  annotate(
    "text",
    x = c(.85, 1.15, 1.85, 2.15),
    y = c(.7 + .2, .7 + .2, .95 + .2, 1.2 + .2),
    label = c("a", "a", "a", "b"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.85, 1.15, 1.85, 2.15),
    # Adjust the y-coordinates to be slightly lower
     y = c(.7 + .1, .7 + .1, .95 + .1, 1.2 + .1),
    # Add your sample sizes here
    label = c("(37)", "(43)", "(35)", "(34)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")


#Combining Plots

# 1. Define a shared theme object
my_shared_theme <- theme_classic() +
  theme(
    axis.title = element_text(face = "bold", size = 18),
    axis.text = element_text(size = 14),
    legend.title = element_text(size = 16),
    legend.text = element_text(size = 14),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA)
  )

# 2. Combine the plots

# Ensure plot1, plot2, and plot3 are already created in your environment
combined_plot = (plot1 | (plot2 / plot3)) +
  plot_layout(widths = c(1.5, 1), guides = 'collect')

# Add the A, B, C tags
combined_with_tags = combined_plot +
  plot_annotation(tag_levels = 'A')

# 3. Apply the shared theme to ALL plots using '&'
final_plot = combined_with_tags &
  my_shared_theme +
  theme(
    plot.tag = element_text(face = "bold", size = 25),
    plot.tag.position = "topleft"
  )


# 3. Save
ggsave(
  filename = "MetabolitesPlot.png",
  plot = final_plot,
  width = 16, height = 10, dpi = 300, bg = "transparent"
)
```